

FR-018 — Opportunity Discovery & Acquisition

FR018 — `sources/acceptance.md`

Masterclass source package snapshot

Functional Requirement	FR-018
Edition	Engineering Learning Academy — Source Package snapshot
Status	**Complete / Frozen / Accepted
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via python scripts/build_masterclass_package.py. Repository documentation remains the source of truth.

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FR-018 — Opportunity Discovery & Acquisition

Status: Complete / Frozen / Accepted

Date: 2026-08-07

Documentation close-out: 2026-08-07

Recommendation: ACCEPT AND FREEZE — FR-018 COMPLETE

Binding product posture: thin Discovery Ingress + channel adapters into frozen Horizon 1A

Engineering Learning Academy: Ready — canonical engineering record = this report; attachable package = [masterclass/FR018/](#) (README.md , MANIFEST.md , regenerable sources/)

Next: FR-019 Recruiter Intelligence on owner request. Do not reopen Horizon 1A. Authenticated browsers / Easy Apply / recruiter workflows are later FRs.

ADR: [ADR-010](#) (Accepted — M1–M4 close-out)

Milestones (historical records): [M0](#) (Accepted), [M1](#) (Accepted), [M2](#) (Accepted — GO WITH REVISED SCOPE), [M3](#) (Accepted — GO WITH REVISED SCOPE), [M4](#) (Accepted — live validation; this freeze).

This document is the **canonical engineering record** for FR-018. Milestone reports remain historical; do not reopen exit criteria without owner request.

1. Executive Summary

FR-018 delivers the **Opportunity Acquisition Framework** for Horizon 1B:

```
OpportunitySource (url | email .eml#job=N)
  ↓
ThinDiscoveryIngress (coordinate only)
  ↓
AcquisitionAdapter (URL | Email → optional URL enrich)
  ↓
ApplicationWorkflowRunner (frozen Horizon 1A)
  ↓
Opportunity (FR-009 SoT)
```

Channel	Status
Owner URL (cic opportunity discover)	Production — SEEK primary; LinkedIn/Indeed attempt/ fail-closed
Owner-saved job-alert .eml (cic opportunity discover-email)	Production — SEEK / LinkedIn / Indeed alerts
Email → URL enrichment	Accepted — card-only alerts reuse <code>UrlAcquisitionAdapter</code>
IMAP / Playwright / recruiter CRM	Out of scope

Live owner validation (LinkedIn alert): `acquired=6 failed=0`; re-run without `--force` → `skipped=6` (definite identity). Full Job Analysis, Assessment, Portfolio Match, and Application Strategy succeeded on enriched ads.

Horizon 1A unchanged. No second orchestrator, Opportunity store, or acquisition Protocol redesign.

2. Engineering Summary

Problem

Horizon 1A can prepare and track applications, but inflow was mostly paste/export. FR-018 scales **lawful** discovery into that loop.

Solution shape

- **Thin ingress** (ADR-010) — resolve → adapter → existing runner; optional definite-identity skip.
- **Reuse FR-008** `AcquisitionAdapter` / `AcquisitionResult`.
- **Channels as adapters** under one FR (Hybrid / Option C), not one mega-FR per board.

Critical refinement (acceptance)

Initial M4 used alert-body text as the JobPosting. Live LinkedIn digests are **discovery cards** (~500 chars: title/company/location). That produced hollow JobAnalysis and flaky Assessment (`Invalid JSON`).

Accepted design: email discovers the job URL; when the alert body is card-only, **enrich via existing** `UrlAcquisitionAdapter` (fail-soft). Provenance remains `source_kind=email`. No parallel acquisition stack.

3. Architecture

Concern	Owner	FR-018 role
Workflow / analyse / assess / strategy	FR-008–FR-011	Unchanged
Opportunity SoT / duplicates	FR-009	Definite-identity skip only
Thin Discovery Ingress	FR-018	Coordinate
URL acquire	FR-018	<code>UrlAcquisitionAdapter</code>
Email discover + optional URL enrich	FR-018	<code>EmailAcquisitionAdapter</code>
Recruiter / outreach / Easy Apply	FR-019+	Out of scope

Invariant: Ingress must not rank, assess, persist Opportunities itself, generate documents, submit, or confirm duplicates.

Package: `career_intelligence.discovery`

CLI: `cic opportunity discover` / `discover-email`

4. Final implementation

Component	Role
<code>OpportunitySource</code>	Transient <code>url</code> or <code>email</code> locator
<code>ThinDiscoveryIngress</code>	Protocol implementation
<code>UrlAcquisitionAdapter</code>	HTTP fetch + HTML extract
<code>EmailAcquisitionAdapter</code>	<code>.eml#job=N</code> parse; optional URL enrich
<code>parse_job_alert_email</code>	Allow-listed senders; SEEK / LinkedIn (<code>/jobs/view</code> + <code>/comm/jobs/view</code>) / Indeed
Provenance asserts	Fail-closed URL and email
TLS	<code>truststore.SSLContext</code> for live fetch (M3)

Email enrichment rule: Prefer URL body when substantially longer **or** when the email lacks JD signals and the URL body has them. Offline fixtures skip enrichment.

5. Live validation evidence

Run	Result
<code>cic opportunity discover-email</code> <code>linkedin_alert.eml --force</code>	acquired=6 skipped=0 failed=0
Enrichment	Full ads via URL adapter ($\approx$ 3.2k–5.5k chars)
Pipeline	JobAnalysis + Assessment + Portfolio Match + Strategy → Opportunities
Re-run without <code>--force</code>	acquired=0 skipped=6 failed=0
Skip reason	Definite identity match on email job URL facets

Unit coverage: `tests/unit/discovery/` (email enrich + ingress regressions).

6. Before vs After

	Initial M4 (pre-acceptance)	Accepted
Email content	Alert card as JobPosting	Card discovers URL; URL adapter supplies JD when possible
LinkedIn live	Parse OK; thin analyse/assess fail/flake	Full pipeline on enriched ads
Architecture	Email-only body path	Reuse canonical URL acquire
Duplicates	Planned definite skip	Confirmed live (skipped=6)

7. Risks

Risk	Mitigation
LinkedIn/Indeed URL fetch intermittent	Fail-soft to card; document limits (M3); paste remains available
Card-only when enrich fails	Honest insufficient strategy / owner paste
Anti-bot / auth walls	No Playwright in FR-018; later FR if needed
Treating email as authoritative JD	Explicit lesson — email is discovery

8. Lessons Learned

1. **Passing unit tests did not demonstrate product readiness** — live owner validation exposed the card-only gap.
 2. **Email alerts are a discovery mechanism**, not an authoritative job description.
 3. **Correct fix was reuse of URL acquisition**, not deeper email HTML parsing.
 4. **Live validation improved design while simplifying architecture** (one acquire path).
 5. **Live validation is mandatory for external integrations.**
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9. Recommendation

ACCEPT AND FREEZE FR-018.

No further implementation for this FR. Future work (authenticated browser, Easy Apply, recruiter intelligence) belongs to **FR-019+**.

10. Final acceptance decision

Decision	Value
FR-018	ACCEPTED
Freeze	FROZEN
Horizon 1A	Unchanged / not reopened
Academy package	masterclass/FR018/
Next FR	FR-019 Recruiter Intelligence (owner request)

FR-018 ACCEPTED. FR-018 FROZEN. READY FOR FR-019.